Since **Jenkins runs inside a Docker container**, you must install **NodeJS inside the Jenkins container**, not on your host machine.

👉 Why?

Because the pipeline executes **inside Jenkins container**, so it must have `node` and `npm`.

# ☑ OPTION 1 (Recommended) — Install via Jenkins NodeJS Plugin

### Step 1: Install Plugin

- Jenkins → Manage Jenkins
- Manage Plugins
- Install **NodeJS Plugin**

### Step 2: Configure NodeJS

- Manage Jenkins → Global Tool Configuration
- Scroll to **NodeJS installations**
- Click **Add NodeJS**

Fill:

- Name: `NodeJS`
- ✅ Install automatically → select latest LTS version

Click Save.

### Step 3: Use in Pipeline

Add this at top:

```
tools {
  nodejs 'NodeJS'
```

```
}
```

Example:

```
pipeline {
  agent any

  tools {
    nodejs 'NodeJS'
  }

  stages {
    stage('Check Node') {
      steps {
        sh 'node -v'
        sh 'npm -v'
      }
    }
  }
}
```

✅ That’s it. Clean & production safe.

# ☑ OPTION 2 (Manual Install Inside Jenkins Container)

Only use if plugin method not allowed.

### Step 1: Enter Jenkins Container

On server:

```
docker ps
```

Find Jenkins container ID.

Then:

```
docker exec -it <jenkins-container-id> bash
```

### Step 2: Install NodeJS

If container is Ubuntu/Debian:

```
apt update
apt install curl -y
curl -fsSL https://deb.nodesource.com/setup_18.x | bash -
apt install -y nodejs
```

Check:

```
node -v
npm -v
```

### Step 3: Restart Jenkins Container

```
docker restart <jenkins-container-id>
```

⚠️ Problem: If container is recreated, Node will be lost.

That's why Plugin method is better.

## 🎯 Interview One-Liner

"Since Jenkins runs inside a Docker container, NodeJS must be installed inside that container, preferably using the Jenkins NodeJS plugin for persistence."